

Ideation Phase

Define the Problem Statements

Date	20 February 2026
Team ID	LTVIP2026TMIDS88068
Project Name	Online Payments Fraud Detection using Machine learning
Maximum Marks	2 Marks

Problem Statement

With the increasing use of online payment systems, financial fraud and unauthorized transactions have also increased. Traditional fraud detection methods are slow and unable to detect new fraud patterns effectively. Therefore, an intelligent system is required to automatically analyze transaction data and identify fraudulent activities in real time.

This project develops an Online Payment Fraud Detection System using Machine Learning to classify transactions as fraud or legitimate and improve digital payment security.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Online banking customer concerned about transaction security.	To perform online payments safely without fraud.	Current systems sometimes fail to detect fraud instantly.	Real-time fraud detection is essential to protect financial data and prevent money loss.	Worried and anxious about unauthorized transactions and financial security.
PS-2	Bank organization managing digital transactions.	To monitor thousands of transactions efficiently.	Manual monitoring is slow and error-prone.	Automated intelligent systems improve fraud detection accuracy.	Pressured to maintain customer trust and security.
PS-3	Online shopper using payment gateways.	To complete purchases quickly and securely.	Cyber fraud and phishing attacks are increasing.	Secure payment systems ensure safe digital transactions.	Unsafe and hesitant while making online payments.